

C.L.A.R.A.

Contextual Locally Aware Robust Agent
Python · FastAPI · React · DeepSeek API · FAISS · SQLite · Ed25519 · MCP · RTX 3050 (4GB VRAM)

WHAT IT IS

CLARA is a personal autonomous AI system built on a simple premise: a capable AI should not need to be invoked for every task. Every component is purpose-built rather than assembled from existing frameworks. The goal was a system that stays present, maintaining context across sessions, running background processes on its own, routing tasks by reasoning complexity rather than keyword matching, and governing and checking its own actions as it goes. It does not just respond. It operates.

SYSTEM FLOW

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Input → EventQueue → Orchestrator → Interpreter → Router → [FAST / CHAT / DELIBERATE]
Response → memorize_episode (background) · All inputs run through the same pipeline. No bypasses.
Every mutating action passes an admissibility gate before it fires.
```

CAPABILITIES

Execution pipeline	FAST / CHAT / DELIBERATE routing by confidence and complexity. Parallel tool batching via asyncio.gather(). FAST auto-escalates to DELIBERATE on failure with full failure context injected.
Governance	A pre-execution admissibility gate scores every mutating action ALLOW / REVIEW / DENY before it fires, recording each to a tamper-evident local ledger. Pluggable local-policy or external-engine adapters. Design partner and reference integration for a third-party governance platform, running signed requests end to end.
Verification & honesty	A deterministic self-grading harness runs twice daily, scoring CLARA's answers against ground truth pulled from source. No model grades a model. Guardrails make her say "I cannot verify that" instead of fabricating. It proves when the agent is wrong, and logs every failure.
Memory system	Semantic episodic retrieval combining recency and cosine similarity. Permanent-fact vault with 0.85 cosine dedup. A verbatim recent-conversation window plus active-discourse tracking for human-like coherence. Temporal grounding, crash-safe persistence, style persistence.
Knowledge base	FAISS RAG indexed on live documentation. Auto-rebuild on source-file changes. Hot-reload without restart. Passive archive injection per request.
Autonomy & ambient	Continuous background loop with scheduled tasks, environment triggers, interrupt handling, crash recovery via SQLite. Consent-gated ambient perception of live context, with grounded recall that reports only what it actually observed.
Tool ecosystem	Native tools (web search, Python REPL, live vision, RAG, ambient recall) plus MCP: Desktop Commander (24 tools) and Markdown document conversion. A semantic tool registry finds the right tool, not just the one you named. Images and documents upload end to end.
Voice, interface & reach	Push-to-talk via Faster-Whisper STT and Kokoro TTS (~200ms latency). Live task board, streaming thought panel, animated vitals over WebSocket. Reachable over Telegram, full two-way, from anywhere.

WHAT SETS IT APART

Most agent systems today are assembled from existing primitives. They work, but the architecture shows. CLARA's memory, knowledge base, autonomy, and reasoning share a single event-driven pipeline because they need to. Context that does not flow between components is not really context. And it does two things most agents cannot: it governs itself, checking every action against a policy before it fires, and it grades itself, proving on a schedule whether its answers were actually correct. Most agents cannot tell you whether yesterday's output was right. CLARA can.

WHERE IT APPLIES

On-device personal AI	Always-on agent with persistent memory. No cloud dependency.
Governed autonomy	For workflows where every action must be authorized before it runs and every claim checked. The gate, the ledger, and the self-audit are built for exactly this.
Enterprise knowledge worker	Autonomous document processing and task orchestration over internal systems.
Developer assistant	Codebase-aware agent with file-system access, tool execution, and project memory.

SCOPE

57 implementation briefs across ~35 source files. Full pipeline in daily use: routing, memory, knowledge base, governance gate, self-verification, ambient awareness, voice, and resource safety all live.

WHAT'S NEXT

- **Streaming migration** — single LLM call with mid-stream tool execution.
- **Proactive ambient (A2)** — salience-gated, unprompted observations.
- **Governance enforcement** — shadow audit to deliberate enforcement.
- **Life context layer** — structured fields for goals, habits, routines.

